WT-RHCA

Networking Exam

November 23, 2025

Important - Network Configuration

First, configure your Host Name, IP Address, Gateway and DNS. Then start the exam.

Network Configuration

Machine 1: Client

- **IP Address:** `192.168.1.50/24`
- **Gateway:** `192.168.1.1`
- **DNS:** `192.168.1.1`
- **Hostname:** `client.example.com`

Machine 2: Server

- **IP Address:** `192.168.1.30/24`
- **Gateway:** `192.168.1.1`
- **DNS:** `192.168.1.1`
- **Hostname:** `server.example.com`

1 Machine 1 Tasks

1. Set multi-user target as your default target. Then, reboot. All the work on this machine should be done in the multi-user target.

2. Find all lines in the file `/usr/share/dict/words` that contain the string **seismic**. Put a copy of all these lines in the file `/root/wordlist`.

3. Create a user **David** with password **tekup**. Configure password ageing for david:

   - Set the mindays to 7
   - Set maxdays to 28
   - Set warndays to 5

4. Create a group called **linuxadm** with GID 5000 and another group called **dba** sharing the GID 5000. Add David as a secondary member to group linuxadm.

5. Change the linuxadm group name to **sysadm** and the GID to 6000. Modify the primary group for David to sysadm.

6. Create a directory `/archive` so that only the owner and the group owner can fully access it. Under `/archive`, create a file named `targets` that contains the list of targets.

7. Create the user **melinda** with uid 4525 with password **"redhat"** and gid 3200.

8. All files that will be created by melinda should have permissions **664** and all directories should have permissions **764**.

9. Copy recursively `/archive` to `/root/archive2`. The owner of `/root/archive2` is david. Its group is sysadm. David, melinda and sysadm can read, write and access to `/root/archive2`. Others can only access `/root/archive2`. Melinda can not delete files on `/root/archive2`. All files that will be created in `/root/archive2` belong to sysadm group.

10. Copy the list of services to a file named `services` under `/root`. Melinda can read and delete this file. The root can read, write and execute this file. The group sysadm can only access this file.

11. Yum packages have been provided at `http://tek-up.de/AppStream` and `http://tek-up.de/BaseOS`. Configure your system to use the two locations as default repos.

12. Create a backup file named `/root/backup.tar.bz2` which contains the content of `/usr/local`.

13. `/data` directory is shared from machine2. Mount the shared directory so that:

    - When a user tries to access the mount point, the directory will be mounted
    - Shared directory should be mounted on `/mnt/data` on this machine

14. As **user10**, build the image named **pdf_conv** using the following:

    - Dockerfile: `https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/master/Dockerfil`
    - Application: `https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/master/pdf_conv`

15. Create a container using the image **pdf_conv**. The container is named **convertor**. Attach volumes `/action/incoming` and `/action/outcoming` (in your machine) to `/data/input` and `/data/output` (in your container) respectively.

16. Create a service `container-pdfconvertor.service` from the container pdfconvertor.

17. Configure a HTTP web server which can be accessed through **port 87**.

18. Write a script that finds all the files owned by **user10** and have size greater than 10KB and less than 10M and store them in `/tmp/find1`.

19. Configure this machine as a client NTP. The server is `server.example.com`.

2 Machine 2 Tasks

1. All passwords should contain digits, uppercases and lowercases. The minimum acceptable size for new passwords is 7.

2. Write a script named `backup.sh` under `/root` which will search files less than 2M from `/usr` and store it in `/root/backup`.

3. As a System Administrator you are responsible to take a backup of your `/etc` directory every night. Build a shell script to take a backup of the `/etc` directory using the tar command. The backup script should be named as `/root/backup.sh`. Schedule this script to run at 11:00 PM every night – except Sundays.

4. Create a 2G swap partition which takes effect automatically at boot-start and should not affect the existing swap partition.

5. Create a volume group called **vgfs** of 160MB created on the `/dev/sdb` disk. The PE size for the volume group should be set at 16MB. Create 2 logical volumes called **ext4vol** and **xfsvol** of size 80MB each and initialise them with the Ext4 and XFS file system types. Ensure that both file systems are persistently defined using their logical volume device filenames. Create mount points `/ext4vol` and `/xfsvol`, mount the file systems, and verify their availability and usage.

6. Extend the **xfsvol** logical volume along with the file system it contains. The new size should be between 157M and 175M.

7. On machine2 (NFS server), create a user account called **user30** with UID 3000. Add the `/home` directory to the list of NFS shares so that it becomes available for remote mount. On machine1 (NFS client), create a user account called **user30** with UID 3000, base directory `/nfshome`, and no user home directory. Establish an indirect map to automount the remote home directory of user30 under `/nfshome`. Observe that the home directory of user30 is automounted under `/nfshome` when you sign in as user30.

8. Create a VDO volume called **vdo1** of logical size 16GB on the sdc disk. Initialise the volume with the XFS file system type, define it for persistence using its device files, create a mount point called `/xfsvdo1`, attach it to the directory structure, and verify its availability and usage.

9. Set the system time zone and configure the system to use NTP.

10. Create a periodic job that appends the current date to the file `~/tracking` every 5 minutes every Sunday and Wednesday between the hours of 3am and 4am. Remove the ability of bob to create cron jobs.

11. Submit a job as **user100** to run the date command at 11:30pm on March 31, 2040, and have the output and any error messages generated redirected to `/tmp/date.out`.

12. Set recommended profile as your tuned profile.

Exam Notes

- Read all questions carefully before starting
- Verify all configurations persist across reboots
- Test services after configuration
- Check file permissions and ownership thoroughly
- Document any issues encountered